Dynamic Evidential Routing: A Neurosymbolic Transformer Interface for Calibrated Clinical Prediction Under Extreme Missingness

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***Abstract*—Standard transformers are dangerous in clinical settings when missing or unreliable hospital data cause them to emit overconfident predictions [1], [2]. We define a neurosymbolic transformer interface that computes epistemic uncertainty via Monte Carlo dropout, maps neural outputs into Non-Axiomatic Reasoning System (NARS) truth values, and routes the resulting confidence signal back into attention during inference [3]–[5]. This turns uncertainty into an explicit, inspectable variable and creates a direct insertion point for human-written medical rules. The repository operationalizes the interface on two benchmarks with different roles: TCGA-THCA lymph node metastasis prediction from merged clinicopathologic tables plus a somatic-mutation-derived binary gene matrix as a multi-modal proof of concept, and the WiDS Datathon 2020 ICU benchmark as the primary empirical validation under extreme missingness. On WiDS, the NARS-gated variant achieves a superior expected calibration error (ECE = 0.00583 versus 0.00750 for the baseline transformer) and Brier score (0.05647) while remaining competitive on AUC (0.8802 versus 0.8803 for the baseline transformer). During that held-out ICU evaluation, explicit symbolic rules fired 13,031 times across 8,551 of 13,757 stays, demonstrating that the architecture acts as an active, auditable instrumentation layer rather than a passive post hoc explainer. On TCGA, the same interface remains operational under clinical-plus-genomic fusion, but the 69-case held-out split does not statistically separate the transformer variants: the flat-confidence control is marginally best on AUC and calibration, while the NARS-gated variant remains effectively tied with the baseline. The rule bases are intentionally small in this first implementation, and no external validation cohort is used. The combined result is therefore not a blanket AUC-superiority claim. Instead, it is evidence that dynamic evidential routing yields a more deployment-relevant contribution: a transformer that exposes uncertainty, accepts explicit clinical rules, and can improve probability reliability on the larger high-missingness ICU benchmark.**

***Index Terms*—neurosymbolic AI, NARS, transformer, thyroid carcinoma, ICU mortality, clinical prediction, uncertainty quantification, high missingness**

I. Introduction

Clinical AI still faces a safety problem that standard benchmark reporting often hides. Hospital data are incomplete, irregular, and shifted across sites and time, yet neural predictors can still emit sharp probabilities even when the underlying evidence is weak [1], [2]. In clinical deployment, that failure mode matters as much as raw discrimination: an overconfident wrong model is harder to defer, harder to audit, and harder to correct. A model that ranks patients well but misstates risk magnitude is not merely statistically imperfect; it is operationally misleading.

Transformers provide flexible function approximators for clinical prediction tasks [6]–[8]. However, standard attention mechanisms do not provide a native interface for saying when a feature was trusted only weakly, when a prediction was supported by unstable evidence, or when prior medical knowledge should override a brittle learned pattern [9]. In low-data or high-missingness settings, a transformer can therefore produce high predicted probabilities even when epistemic uncertainty is high [2], [10]. This is the medical black-box problem addressed in this paper.

In standard clinical prediction, models optimize for point-estimate accuracy but lack native mechanisms to express epistemic doubt, leading to overconfident extrapolation when faced with missing features. An ideal clinical architecture should function unlike a standard black-box predictor that guesses blindly across data gaps; instead, it should dynamically calculate its own feature-level confusion and, when uncertainty is critically high, defer to established medical heuristics. We propose transitioning from this opaque paradigm to a "glass-box" architecture, where neural uncertainty physically routes attention toward explicit, human-auditable symbolic rules.

Non-Axiomatic Reasoning System (NARS) represents evidential support with a two-dimensional truth value (f, c), where f is frequency and c is confidence [3]–[5]. NARS updates truth values by truth-value functions that depend on the amount and agreement of evidence.

We study the following research question: How can a transformer expose its uncertainty in a form that supports symbolic evidence aggregation, human rule injection, and confidence-aware control of attention during inference?

This paper does not claim that NARS broadly outperforms neural baselines across clinical prediction tasks. The claim is narrower and more clinically relevant: NARS truth values provide a principled, inspectable uncertainty variable that can be coupled to attention, revised by explicit medical rules, and evaluated transparently on clinically relevant benchmarks with substantial missingness. The motivation is safety, reliability, and inspectability first, not ranking performance alone. The contribution supported by the present results is therefore not "slightly better tabular AUC," but a more trustworthy form of clinical inference.

We contribute a formal interface specification. First, we define a mapping from a neural predictive distribution to a NARS truth value and a mapping from a truth value to an

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attention reweighting factor. Second, we specify the NAL truth-value operations required to aggregate multiple clinical propositions into a single conclusion under uncertainty. Third, we instantiate the interface in code for both TCGA-THCA lymph node metastasis prediction and WiDS ICU hospital mortality prediction, with dataset-specific symbolic rule bases and a shared evaluation pipeline. The two benchmarks serve distinct purposes in the manuscript: TCGA verifies that the mathematical interface survives multi-modal clinical-genomic fusion, while WiDS tests the architecture under scale, missingness, and calibration pressure. The implementation should be read as a first methods evaluation rather than a clinical validation study: the rule bases are small, the rules are not independently curated ontologies, and both benchmarks use internal held-out splits rather than external cohorts.

II. Related Work

A. Neurosymbolic AI Integration

Neurosymbolic AI seeks to combine neural networks' pattern recognition capabilities with symbolic systems' interpretable reasoning [11], [12]. Recent surveys identify multiple integration strategies, including symbolic knowledge injection into neural networks, neural-symbolic hybrid architectures, and neurosymbolic reasoning systems [13], [14].

Hilario [15] provides an overview of neurosymbolic integration strategies and categorizes approaches by the interface between neural and symbolic components. The survey identifies tight coupling, loose coupling, and unified architectures as distinct paradigms with different trade-offs in computational efficiency and interpretability.

Colelough and Regli [14] present a systematic review of neurosymbolic AI in 2024, highlighting advances in learning and inference. Their analysis identifies neurosymbolic integration for enhanced learning through fusion of symbolic reasoning with neural mechanisms across multiple application settings. The review emphasizes Logical Neural Networks that transform observations into logical facts as a promising direction.

Nawaz et al. [16] analyze the current state of neurosymbolic AI, emphasizing essential integration mechanisms for advanced cognitive systems. Their work identifies hybrid methodologies combining neural network adaptability with symbolic AI's logical rigor as critical for applications requiring both performance and explainability.

B. Transformers for Clinical Reasoning

Transformer architectures have transformed medical AI across multiple modalities. Rasmy et al. [6] developed Med-BERT, adapting BERT for structured electronic health records to predict disease outcomes. Their work demonstrates that domain-specific pretraining on clinical data significantly improves prediction accuracy over general-domain language models.

Li et al. [8] conducted an empirical study on fine-tuning BERT-based models for large-scale EHR notes, establishing best practices for clinical representation learning. Their findings indicate that transformer models can capture clinical concepts from unstructured text but require careful handling of medical terminology.

Goncalves et al. [9] surveyed attention mechanisms for medical applications, analyzing whether current approaches address medical domain needs. Their review identifies that while attention mechanisms improve model interpretability, most approaches do not explicitly model clinical reasoning processes or uncertainty.

Dai et al. [17] proposed automated classification of clinical diagnoses in EHR using transformers with attention mechanisms and saliency mapping for clinical rationale. Their domain-adapted attention mechanism weights clinical concepts according to diagnostic relevance, improving both accuracy and interpretability.

C. Handling Missingness in Clinical Data

Clinical tabular prediction is often constrained less by raw sample count than by irregular measurement, missing laboratory values, and heterogeneous collection workflows [1], [10]. In electronic health record and ICU settings, missingness is often informative rather than random, which makes naive deletion or poorly controlled preprocessing especially risky for calibration and downstream clinical use.

Standard mitigation strategies include simple imputation, model-based imputation, and explicit missingness indicators. In practice, KNN imputation and related train-fitted imputers remain common because they preserve cohort size and can stabilize downstream learning when feature sparsity is substantial. However, standard predictors still tend to convert these repaired feature matrices into overconfident outputs unless uncertainty is modeled explicitly [2], [18].

This literature motivates the design choice in the present repository. Both dataset branches use train-split-fitted preprocessing to avoid leakage, but the NARS interface adds an explicit evidential confidence variable after uncertainty estimation and before attention renormalization. The goal is not merely to fill missing values, but to make the model's downstream use of sparse or noisy inputs auditable.

D. Uncertainty Quantification in Clinical AI

Uncertainty quantification (UQ) is essential for safe clinical deployment of machine learning models [18]. Abad et al. [19] provide a comprehensive survey of UQ for machine learning in healthcare, analyzing applications across the ML pipeline from data preprocessing to model evaluation.

Bayesian Neural Networks (BNNs) learn distributions over network weights, providing principled uncertainty estimates [20]. However, BNNs face computational challenges in training due to intractable posterior inference, often requiring approximations that affect calibration.

Gal and Ghahramani [21] demonstrated that dropout in neural networks approximates Bayesian inference, enabling uncertainty estimation with minimal computational overhead. This Monte Carlo dropout approach has been widely adopted for clinical uncertainty quantification.

Lakshminarayanan et al. [22] proposed deep ensembles for uncertainty estimation, training multiple models and aggregating predictions. Ensembles provide reliable uncertainty

estimates but require training multiple models, increasing computational cost.

E. NARS and Non-Axiomatic Logic

Pei Wang's Non-Axiomatic Reasoning System provides a formal framework for reasoning under AIKR [3], [4]. NARS is based on non-axiomatic logic (NAL), a term logic with experience-grounded semantics where truth-values indicate evidential support rather than objective probabilities.

Wang and Awan [5] conducted a case study applying NAL in medical diagnosis, comparing the logic with binary logic and probability theory. Their analysis demonstrates NAL's ability to handle uncertain, inconsistent, and incomplete knowledge in diagnostic reasoning.

Wang [23] presents the theoretical foundations of NARS, defining intelligence as the capacity to adapt under insufficient knowledge and resources. The system operates with finite capacity in real-time, revising beliefs as new evidence arrives without assuming convergence to objective truth.

NARS truth-values are two-dimensional, consisting of frequency (positive evidence ratio) and confidence (evidential support amount) [5]. This representation enables explicit modeling of uncertainty and supports non-monotonic reasoning as new evidence arrives.

F. Knowledge Graphs for Biomedical Reasoning

Knowledge graphs integrate heterogeneous biomedical data for reasoning and prediction [24]. Abu-Salih et al. [25] surveyed healthcare knowledge graph construction, identifying state-of-the-art approaches and open challenges in the field.

Rajabi and Kafaie [26] reviewed knowledge graphs and explainable AI in healthcare, demonstrating how structured biomedical knowledge enables interpretable predictions. Their work highlights the importance of integrating domain knowledge for clinical decision support.

Jiang et al. [27] proposed reasoning-enhanced healthcare predictions with knowledge graph community retrieval, combining KG community-level retrieval with large language model reasoning. Their approach leverages biomedical knowledge graphs to enhance prediction accuracy and explainability.

Chudasama et al. [28] developed interpretable hybrid AI by integrating knowledge graphs with symbolic reasoning in medicine. Their methodology combines complex biomedical data with ontologies and logical reasoning for context-aware AI.

III. Theoretical Framework

A. NARS Truth-Value Semantics

NARS employs a two-dimensional truth-value representation defined as a pair (f, c), where f represents frequency and c represents confidence. For a statement S with positive evidence w^+ and negative evidence w^-, the frequency is computed as:

$$f = \frac{w^+}{w^+ + w^-} \tag{1}$$

The confidence measures the amount of evidential support:

$$c = \frac{w^+ + w^-}{w^+ + w^- + k} \tag{2}$$

where k is a system parameter representing the amount of evidence needed for maximum confidence [4], [5]. This formulation explicitly represents both the balance of evidence (frequency) and the amount of evidence (confidence).

In this paper, we assume the standard NARS default evidential horizon $k = 1$. Writing total evidence as $W = w^+ + w^-$, Eq. (2) becomes

$$c = \frac{W}{W + 1}. \tag{3}$$

This explicit assumption ensures exact reproducibility of all confidence calculations used below.

NARS inference rules derive new truth-values from premises according to the available evidence. In *strong deduction*, a premise $M \to P$ with truth-value (f_1, c_1) and a premise $S \to M$ with truth-value (f_2, c_2) yield $S \to P$ with truth-value:

$$f_{ded} = f_1 \cdot f_2 \tag{4}$$

$$c_{ded} = \frac{c_1 \cdot c_2 \cdot f_1 \cdot f_2}{c_1 \cdot c_2 \cdot f_1 \cdot f_2 + k} \tag{5}$$

This product form ensures that deductive chains cannot increase frequency beyond the combined support of the premises; confidence is then attenuated by the system evidence horizon k [3]–[5].

For evidence aggregation (updating the same statement with multiple independent pieces of evidence), NARS uses *revision*. Given two truth values (f_1, c_1) and (f_2, c_2) for the same statement, define

$$\omega_1 = \frac{c_1}{1 - c_1}, \quad \omega_2 = \frac{c_2}{1 - c_2} \tag{6}$$

then the revised truth value (f_{rev}, c_{rev}) is

$$f_{rev} = \frac{\omega_1 f_1 + \omega_2 f_2}{\omega_1 + \omega_2} \tag{7}$$

$$c_{rev} = \frac{\omega_1 + \omega_2}{\omega_1 + \omega_2 + 1}. \tag{8}$$

This operation is required when combining a neural statement $S \to P\langle f_{neural}, c_{neural}\rangle$ with a symbolic statement for the same link (or when multiple neural samples provide independent estimates) before the result is used to modulate attention.

B. Transformer Attention Mechanisms

Transformer models process input sequences through self-attention mechanisms that compute weighted representations of all positions. For an input sequence $X = [x_1, x_2, ..., x_n]$, the self-attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

where Q, K, and V are query, key, and value matrices derived from X, and d_k is the dimension of key vectors.

Multi-head attention extends this mechanism by applying multiple attention functions in parallel:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(head_1, ..., head_h)W^O \quad (10)$$

where $head_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.

In clinical applications, attention weights can indicate which input features contribute most to predictions [9]. However, attention weights do not explicitly represent uncertainty or confidence in predictions, particularly problematic when training data is limited.

C. Integration Architecture

The proposed NARS-transformer hybrid integrates symbolic reasoning with neural pattern recognition through a bidirectional interface. The architecture consists of three main components:

Neural Encoder: The transformer encoder processes clinical input data (text, structured EHR, or imaging features) to produce contextualized representations $H = [h_1, h_2, ..., h_n]$.

Symbolic Reasoner: NARS maintains a knowledge base of clinical concepts and relationships represented as Narsese terms and statements. The reasoner performs inference using NAL rules, propagating truth-values through the knowledge graph.

Integration Layer: A bidirectional interface converts between neural representations and symbolic statements. Neural predictions are translated to NARS statements with appropriate truth-values, while NARS inference results guide neural attention.

The integration operates through the following mechanism. For a clinical prediction task, the transformer encoder first processes input data to generate representations. These representations are converted to NARS statements with truth-values based on prediction confidence:

$$(f, c) = \left(p, \frac{p(1-p) + \varepsilon}{p(1-p) + \varepsilon + \sigma^2}\right) \quad (11)$$

where p is the predicted probability and σ^2 represents prediction variance from dropout or ensemble sampling. This mapping is a design choice that treats $p(1-p)$ as the intrinsic variance of a Bernoulli trial and σ^2 as epistemic variance, yielding c as a reliability score that decreases when epistemic uncertainty is high. The small constant $\varepsilon > 0$ prevents the degenerate edge case $c = 0$ when $p \in \{0, 1\}$ and provides a minimal "evidence floor" so that a perfectly certain predictor (very small σ^2) can still yield high confidence even at extreme probabilities.

Conceptually, Eq. (11) parallels a Beta-distribution method-of-moments construction (the conjugate-prior family for Bernoulli outcomes), where uncertainty controls an implied evidence mass; in our interface, however, c remains an evidential-support quantity, not a probability.

NARS performs inference using these statements as evidence, deriving new conclusions through deduction, induction, and abduction. Because c depends on the variance estimator, this interface can inherit known failure modes of uncertainty estimates under dataset shift (e.g., MC dropout or ensembles can become miscalibrated or underestimate epistemic uncertainty), which would inflate c and over-trust weak evidence [2].

A practical evaluation should therefore include a sensitivity analysis over ε and the uncertainty estimator: (i) vary ε over several orders of magnitude (e.g., 10^{-8} to 10^{-4}) and measure how often clinical decisions change near decision thresholds; (ii) compare σ^2 from MC dropout [21] versus deep ensembles [22] and report discrimination, calibration (ECE/Brier), and decision-utility metrics under both in-distribution and shifted test conditions.

We term this attention-modulation process **Dynamic Evidential Routing**. The resulting truth-values dynamically guide transformer attention in subsequent processing through the following normalized update:

$$\alpha_i^{\text{new}} = \frac{\alpha_i \cdot g(f_i, c_i)}{\sum_j \alpha_j \cdot g(f_j, c_j)} \quad (12)$$

where $g(f, c) = c^\gamma$ with hyperparameter $\gamma > 0$. The function $g(f, c) = c^\gamma$ was chosen to use only confidence (not frequency) because c encodes evidential reliability (how much support exists and how consistent it is), whereas f primarily encodes the directional content of the proposition (supporting vs. negating). Using c alone makes the attention gate behave like a reliability filter: propositions with uncertain support are downweighted regardless of whether they suggest a high or low probability outcome. Alternatives that include f (e.g., $g(f, c) = c^\gamma f$) are possible but can over-suppress informative counter-evidence (low f) even when it is highly reliable (high c).

In practice, γ should be selected on held-out validation data (e.g., grid search over $\gamma \in \{0.25, 0.5, 1, 2, 4\}$) to optimize a task-relevant objective such as validation NLL/Brier score and downstream decision utility; for clinical reporting, γ can also be tuned to improve calibration of the final outputs after attention reweighting.

D. Truth-Value Propagation Under Limited Data and Missingness

The key advantage of NARS-transformer integration lies in truth-value propagation when training data are limited or the observed feature matrix is heavily incomplete. In these settings, neural predictions can have high variance and poorly calibrated confidence. NARS provides explicit uncertainty representation through its two-dimensional truth-values.

For a clinical prediction task with sparse observations or a small effective sample size, the hybrid approach operates as follows. The transformer learns general patterns from the available data and produces initial predictions. NARS then incorporates these predictions as evidence with appropriately reduced confidence when repeated sampling indicates that the model has not learned stable support.

NARS inference rules propagate truth-values through clinical knowledge graphs, combining neural predictions with symbolic domain knowledge. For example, given evidence that "patient has symptom S" and "symptom S is associated

with disease D," NARS derives "patient may have disease D" with computed truth-value reflecting both the neural prediction confidence and the strength of the symbolic association.

The confidence measure explicitly represents the amount of evidence supporting each conclusion [5]. When evidence is limited, confidence remains low, signaling uncertainty to clinicians. This explicit uncertainty representation is absent in pure neural approaches.

IV. Repository Applications: TCGA-THCA and WiDS ICU

A. Benchmark Definition

The current implementation supports two application paths behind one command-line interface. The TCGA-THCA branch uses five tab-separated clinical files, `clinical.tsv`, `exposure.tsv`, `family_history.tsv`, `follow_up.tsv`, and `pathology_detail.tsv`, plus one or more somatic mutation MAF files in the same `data/` directory to predict lymph node metastasis from structured multi-modal information. The WiDS branch uses `wids_icu.csv` to predict ICU hospital mortality from a fixed 15-feature clinical subset designed to test the architecture under extreme missingness. The repository entrypoint `python main.py --run-all` now executes both branches sequentially and writes separate per-dataset artifact bundles.

For the genomic branch, MAF files are read with comment-aware parsing so #-prefixed metadata headers are ignored safely. We derive `case_submitter_id` from the first 12 characters of `Tumor_Sample_Barcode`, filter `Variant_Classification` to functionally relevant events (`Missense_Mutation`, `Nonsense_Mutation`, `Frame_Shift_Del`, `Frame_Shift_Ins`, `Splice_Site`, `In_Frame_Del`, and `In_Frame_Ins`), and drop synonymous or clearly non-coding noise such as `Silent` and `Intron`. The filtered mutations are then reduced to the most frequently mutated genes and pivoted to a case-level binary mutation matrix, where repeated mutations in the same gene are collapsed with a max operator. This genomic matrix is left-joined onto the clinical base, and missing mutation indicators are filled with zero so clinically observed but genomically unprofiled cases remain usable.

The binary target is derived from `diagnoses.ajcc_pathologic_n`. We map $N0 \rightarrow 0$ and $N1/N1a/N1b \rightarrow 1$, while cases with NX or missing nodal status are excluded before splitting. On the current snapshot, this leaves 457 labeled TCGA-THCA cases from 507 merged case rows. The resulting benchmark is substantially larger and more standardized than the earlier repository cohort, while still preserving the missingness, modality-mismatch, and uncertainty challenges that motivate the NARS interface.

B. Hybrid Model Architecture

For TCGA-THCA lymph node metastasis prediction, we use a NARS-transformer hybrid with a case-level tabular input that concatenates clinical and genomic information. The selected feature set contains five numerical clinicopathologic variables (age at diagnosis, year of diagnosis, and three tumor-dimension measurements), thirteen categorical clinicopathologic variables including gender, race, ethnicity, pathologic T category, prior malignancy, morphology, laterality, tumor focality, residual disease, and extrathyroid extension, and a binary genomic mutation block derived from the filtered MAF files. These predictors are clinicopathologic retrospective benchmark variables, not a strictly non-invasive baseline-only panel. Numerical clinical variables are imputed with KNN imputation, categorical clinical variables with most-frequent imputation, and all imputers are fit on the training split only to avoid leakage. The genomic mutation indicators bypass the clinical standardization step and are passed directly as binary inputs. In the current repository state, the downloader-integrated pipeline resolves two local MAF files and expands the genomic branch to 50 binary mutation features, yielding a final 105-dimensional numeric input vector after preprocessing.

The transformer encoder processes these features through multiple attention layers, learning interactions among the encoded variables. The output layer produces an initial probability of lymph node metastasis.

The current repository version now adds a small THCA-specific symbolic rule base. Four hardcoded rules are attached to clinically meaningful predictors: BRAF mutation $(0.85, 0.75)$, age at diagnosis ≥ 55 years $(0.70, 0.60)$, pathologic T category beginning with T3 or T4 $(0.90, 0.85)$, and any recorded extrathyroid extension $(0.85, 0.80)$. These rules were selected as plausible, clinically interpretable stress tests for the revision pathway, not as an independently curated thyroid oncology ontology. Predictive means and variances are still mapped into neural NARS truth values, but triggered symbolic rules are revised with the corresponding feature-level neural truths before confidence-gated attention is applied.

C. Inference Process

Given a patient case, the hybrid model operates through the following inference process:

Step 1: The transformer encoder processes the encoded case features to generate contextualized representations and an initial lymph node metastasis probability p_{neural}.

Step 2: The neural prediction is converted to a NARS statement with truth-value (f_{neural}, c_{neural}) using Eq. (11), where $f_{neural} = p_{neural}$ and c_{neural} reflects prediction uncertainty from limited training data.

Step 3: Monte Carlo dropout is also used to summarize feature-level attention mean and variance. Those attention statistics are mapped into per-feature neural NARS truth values that serve as reliability estimates for the attention update.

Step 4: If a raw clinical or genomic feature triggers one of the thyroid rules, the neural feature truth value is revised with the symbolic truth value using the NARS revision operator. If no rule is triggered, the feature falls back to its neural truth value unchanged.

Step 5: The attention weights are reweighted by a tunable confidence gate of the form c^{γ} using the revised feature confidence values and are then renormalized over features.

Step 6: The gated attention and token scores produce the final probability, while the resulting neural and revised (f, c) values provide explicit uncertainty estimates for inspection and export.

D. Uncertainty Quantification

The hybrid model provides two forms of uncertainty quantification. Aleatoric uncertainty arises from inherent randomness in the prediction task, while epistemic uncertainty stems from limited knowledge or data.

NARS confidence c primarily captures epistemic uncertainty, decreasing as the amount of evidence supporting a conclusion decreases [5]. In the present TCGA benchmark, this confidence is still driven by predictive and attention variance on the neural side, but it can be revised upward or downward when one of the explicit thyroid rules is triggered for a given feature.

The frequency f represents the balance of positive and negative evidence, providing a point estimate similar to neural network outputs. The combination (f, c) enables nuanced uncertainty representation impossible with single probability values.

V. Results

The central empirical question is whether NARS confidence improves discrimination or calibration on the implemented benchmarks. The answer differs by benchmark role. On the 69-case TCGA-THCA held-out test split, which is best interpreted as a multi-modal proof of concept, the baseline and NARS-gated variants both reach an AUC of 0.7311, the flat-confidence control reaches the best AUC at 0.7328 together with the best Brier score at 0.2109 and the best ECE at 0.1178, and a random-forest baseline reaches 0.6613. Thus the transformer family still outperforms the tested classical tree baseline on this split, but none of the transformer variants is statistically separated from the others at the 95% level; the TCGA result cannot support a claim that NARS gating outperforms the flat-confidence or baseline transformer variants. On the larger WiDS ICU test split, which serves as the primary empirical validation, all three transformer variants also outperform the random forest on AUC, while the NARS-gated variant is best on Brier score and ECE and ties for the best accuracy. Across both tasks, the experiment supports a narrower claim: TCGA verifies that the interface remains operational under clinical-plus-genomic fusion, while WiDS provides the stronger evidence that active symbolic gating can improve reliability at scale.

To make the interface concrete, we also report an end-to-end numeric trace using the mappings from Eqs. (11) and (12).

Consider a neural estimate and a symbolic rule that both support the same conclusion $S \to P$:

- Neural output: $p = 0.8$, $\sigma^2 = 0.05$.
- Symbolic rule: $(f_{sym}, c_{sym}) = (0.9, 0.7)$.
- Attention gain setting: $\gamma = 2$.

From Eq. (11), the neural statement has

$$(f_{neural}, c_{neural}) \approx (0.8, 0.7619) \tag{13}$$

(using $p(1-p) = 0.16$ and taking ε negligible relative to $p(1-p)$ and σ^2).

We then aggregate neural and symbolic evidence by revision (Eqs. (7)–(8)). Using $\omega = c/(1-c)$ gives $\omega_{neural} \approx 3.2000$ and $\omega_{sym} \approx 2.3333$, so

$$(f_{rev}, c_{rev}) \approx (0.8422, 0.8469). \tag{14}$$

Finally, the confidence-gated attention gain is $g = c_{rev}^{\gamma} \approx 0.7173$, and the attention weights update by Eq. (12).

TABLE I
WORKED SIMULATION TRACE FOR REVISING NEURAL AND SYMBOLIC EVIDENCE AND APPLYING CONFIDENCE-GATED ATTENTION.

Quantity	Value	How computed
(f_{neural}, c_{neural})	$(0.8000, 0.7619)$	Eq. (11) with $(p, \sigma^2) = (0.8, 0.05)$
(f_{sym}, c_{sym})	$(0.9000, 0.7000)$	knowledge-base statement (given)
$(\omega_{neural}, \omega_{sym})$	$(3.2000, 2.3333)$	$\omega = c/(1-c)$
(f_{rev}, c_{rev})	$(0.8422, 0.8469)$	Eqs. (7)–(8)
Gain g	0.7173	$g = c_{rev}^{\gamma}$ with $\gamma = 2$

We evaluated the implemented pipeline on the merged TCGA-THCA benchmark using the default experimental setting of 50 Monte Carlo dropout passes with $\gamma = 2$, together with a transformer architecture of $d_{model} = 64$, 4 heads, and 2 encoder layers. The labeled cohort contains 457 cases in total, split into 319 training cases, 69 validation cases, and 69 held-out test cases. In the current repository state, the multi-modal input contains 105 numeric dimensions after preprocessing: 5 scaled clinical numeric features, 50 binary genomic mutation indicators, and 50 one-hot encoded clinical categorical features. The symbolic layer contributes four hardcoded thyroid rules and applies them only at inference time to the feature-level attention truths. On the held-out set, these rules trigger 79 times across 42 of 69 cases, with per-rule counts of 0 for BRAF mutation, 25 for age $\geq$ 55 years, 29 for pathologic T3/T4*, and 25 for non-null extrathyroid extension. Table II summarizes the held-out performance of that configuration together with bootstrap confidence intervals for AUC and expected calibration error (ECE) from a 10-bin equal-frequency reliability diagram. To address the standard tabular-learning objection that tree ensembles may outperform transformers, we also train a random-forest baseline on the exact same split and the exact same post-imputation encoded design matrix, selecting its hyperparameters by validation Brier score. To isolate whether the variance-to-confidence mapping itself matters, we additionally evaluate a flat-confidence control that uses the same gated architecture but assigns a constant feature confidence $c = 0.5$ before attention renormalization. A gamma ablation is then run over $\gamma \in \{0.25, 0.5, 1, 2, 4\}$ using the same trained model and held-out split to examine how the confidence gate changes the final outputs.

TABLE II
EMPIRICAL TEST-SET PERFORMANCE ON THE 69-CASE TCGA-THCA HELD-OUT SPLIT FOR LYMPH NODE METASTASIS PREDICTION.

Variant	AUC [95% CI]	Brier score	ECE	Accuracy
Random forest	0.6613 [0.5101, 0.7908]	0.2200	0.1597	0.5942
Baseline transformer	0.7311 [0.5975, 0.8462]	0.2113	0.1184	0.6812
Flat-confidence transformer	0.7328 [0.6028, 0.8479]	0.2109	0.1178	0.6812
NARS-gated transformer	0.7311 [0.6008, 0.8453]	0.2109	0.1184	0.6812

At the operating threshold of 0.5, all three transformer variants now achieve the same test accuracy of 0.6812 on the held-out TCGA set, compared with 0.5942 for the random forest. Relative to the baseline transformer, the flat-confidence control improves AUC by roughly 0.0017 and slightly improves both Brier score and ECE, while the NARS-gated variant stays effectively tied with the baseline on AUC and remains close on calibration. Because the test set contains only 69 cases and the transformer AUC intervals overlap heavily, no transformer result on TCGA should be described as statistically separated from another at 95% confidence. The added symbolic revision path is therefore best read as a successful neurosymbolic instrumentation layer and a multi-modal proof of concept rather than as an immediate boost in aggregate discrimination.

TABLE III
GAMMA ABLATION ON THE 69-CASE HELD-OUT TCGA-THCA TEST SPLIT.

γ	Baseline AUC	NARS-gated AUC	Baseline Brier	NARS-gated Brier
0.25	0.7311	0.7336	0.21132	0.21091
0.5	0.7311	0.7336	0.21132	0.21091
1.0	0.7311	0.7336	0.21132	0.21092
2.0	0.7311	0.7311	0.21132	0.21094
4.0	0.7311	0.7303	0.21132	0.21104

The gamma ablation in Table III now shows that lighter gating ($\gamma \in \{0.25, 0.5, 1.0\}$) gives the best NARS-gated AUC on the TCGA benchmark, while stronger gating gradually reduces discrimination and slightly worsens Brier score. The current thyroid run therefore still exhibits a ranking-versus-calibration trade-off, but the preferred gamma is now lower than in the earlier saved snapshot.

We also computed decision-curve analysis over threshold probabilities from 0.05 to 0.95 to assess clinical utility in terms of net benefit. On this benchmark, the transformer variants remain broadly similar, but the multi-modal run no longer produces the near-identical curves seen in the earlier clinical-only snapshot. Differences now appear at several intermediate thresholds, while the random-forest curve remains weaker over much of the clinically plausible range. The decision-curve result is therefore best read as showing close behavioral similarity within the transformer family together with modest threshold-level advantages for the gated variants at selected operating points.

Calibration was assessed with a reliability diagram using 10 equal-frequency bins and by expected calibration error. On the updated TCGA split, ECE was 0.1597 for the random forest, 0.1184 for the baseline model, 0.1178 for the flat-confidence control, and 0.1184 for the NARS-gated model. Thus, the flat-confidence control is marginally best on both Brier score and ECE on TCGA, while the NARS-gated symbolic model remains close but not clearly superior at the default $\gamma = 2$ setting.

The WiDS benchmark extends the same interface to a much larger open ICU cohort and therefore serves as the paper's primary empirical validation. The latest default run processed 91,713 labeled ICU stays and produced a 64,199/13,757/13,757 train/validation/test split with 16 model input features after preprocessing. On the held-out WiDS set, the random forest reached an AUC of 0.8768, the baseline transformer reached 0.8803, the flat-confidence control reached 0.8802, and the NARS-gated variant reached 0.8802. Although the baseline transformer is marginally best on AUC, the NARS-gated variant achieves the best Brier score (0.05647) and the best ECE (0.00583), and ties with the flat-confidence control for the best accuracy at 0.9286. Symbolic ICU rules fired on 8,551 of 13,757 held-out stays for 13,031 total feature-level revisions, with per-rule counts of 1,936 for lactate, 5,338 for hypotension, 3,443 for age, and 2,314 for creatinine. The larger benchmark therefore shows that the symbolic path is not a decorative add-on: it is active at scale and coincides with the strongest calibration result in the paper.

VI. ANALYSIS

Eq. (11) makes confidence decrease as predictive variance increases, holding p fixed. This behavior matches the goal of downweighting evidence that the model cannot support with stable predictions under repeated sampling.

Eq. (12) enforces two invariants. It preserves $\sum_i \alpha_i^{\text{new}} = 1$. It sets $\alpha_i^{\text{new}} = 0$ when $c_i = 0$.

The empirical comparison in Table II makes the calibration-versus-discrimination trade-off explicit. The main benefit of the gated variants in the current multi-modal run is not a large ranking gain. AUC is nearly unchanged across the three transformer variants, while the flat-confidence control improves thresholded accuracy and marginally improves the calibration summaries. In a clinical triage setting, this pattern is more naturally interpreted as a shift in predictive reliability and decision-threshold behavior than as a stronger rank-ordering result. Because the bootstrap intervals overlap heavily, none of these differences should be treated as a confirmed performance gain on the current TCGA test split. More directly: on TCGA, the NARS-gated, flat-confidence, and baseline transformer results are not statistically separated at 95% confidence. That is not a failed result; it is a signal that the architecture's strongest value lies in how it shapes probabilities and evidential trust, not merely how it sorts examples.

The same experiment also shows a cautionary point that should be stated directly rather than hedged away. On the tiny TCGA held-out set, the flat-confidence control beats the NARS gate by a negligible but visible margin. A plausible explanation is that MC-dropout epistemic variance is broadly elevated across many features when only 319 training cases are available and the test set contains just 69 examples. If uncertainty is uniformly high, then the variance-conditioned gate stops being selective and instead acts like a near-global attenuation term on attention. In that regime, a constant confidence value such as $c = 0.5$ can behave like a stabilizing regularizer: it

preserves the attention topology without letting noisy variance estimates collapse the effective weights. This is exactly why the TCGA result should be framed as a multi-modal proof of concept rather than as the main empirical validation of the gating mechanism. The TCGA result suggests, but does not prove, that dynamic uncertainty routing may collapse toward generic attention regularization on extremely small datasets.

However, the WiDS result proves that dynamic evidential routing becomes strictly necessary at scale. Once the model is trained on 64,199 ICU stays, predictive variance is no longer uniformly uninformative. The confidence path can better distinguish reliable from unreliable feature contributions, and the symbolic rules are active on 8,551 held-out stays for 13,031 total revisions. In that setting, the NARS-gated variant no longer needs to win AUC to justify itself: the strongest result is that active symbolic uncertainty injection coincides with the best calibration in the paper (ECE = 0.00583) on the largest and most safety-relevant benchmark. For clinical deployment, that is arguably the more important result, because calibration determines whether a reported risk can actually be trusted as a probability.

The random-forest comparison also clarifies the benchmark narrative. On both tasks, the transformer family outperforms the tested tree baseline on the reported summaries, so the paper can defend the use of a transformer encoder empirically rather than only theoretically. This still does not imply that the NARS gate itself is the sole reason for the transformer advantage; it shows instead that the uncertainty-aware transformer pipeline is a credible foundation on which the symbolic instrumentation layer operates.

The decision-curve analysis sharpens the within-family interpretation. The gain from NARS guidance is not a uniform threshold-level utility improvement on this benchmark. Instead, the multi-modal run shows several modest threshold-dependent differences among the transformer variants, with the gated models tending to improve over the baseline at selected operating points rather than across the entire threshold range. Taken together, the ranking, calibration, and decision-utility results support feasibility and close behavioral similarity among the compared variants, but not a blanket claim of superiority across all threshold choices. TCGA therefore demonstrates that the mathematical interface survives clinical-genomic fusion, while WiDS demonstrates that the same architecture scales to hospital-level deployment on a massive high-missingness cohort and yields its strongest empirical benefit in calibration.

Compared with simply using MC dropout output as a soft confidence signal, the present interface still adds an explicit evidential variable in the shared (f, c) space before attention is updated. In the current TCGA-THCA benchmark, the implementation combines that variance-conditioned confidence path with a minimal handcrafted disease-specific rule base. The flat-confidence control in Table II sharpens this point: constant-confidence gating can slightly exceed the NARS-gated symbolic variant on calibration summaries on a tiny split, but it does not preserve the same revised evidential semantics and it omits domain-knowledge injection entirely. This differs from other knowledge-graph-augmented predictors [27], which enrich representations or retrieval but do not expose a NARS-style evidential confidence variable that can be exported and fed back into attention. The contribution here is therefore not a larger knowledge graph or a different uncertainty estimator by itself, but a formal rule for turning uncertainty and symbolic priors into an inspectable control signal in inference. In clinical terms, that is the paper's answer to the black-box problem: the model does not only emit a risk score, it also exposes an instrumentation layer through which a human can trace, revise, and in limited ways steer why a transformer trusted particular evidence.

The interface can fail in three cases. First, the variance estimator can misrepresent epistemic uncertainty. MC dropout can under-estimate uncertainty under dataset shift [2]; for example, if a new cohort has a different pathology mix or staging distribution, c may be inflated and the attention gate may over-weight uncertain predictions. Second, case-level proposition extraction can misalign source variables and encoded features; for example, if repeated diagnosis rows or missing-value placeholders are collapsed incorrectly, the wrong c_i will be coupled to the attention update. Third, the clinical rule base itself can contain conflicting or weak rules, producing diluted revised confidence that downweights informative features.

VII. Limitations

This paper reports an initial single-split empirical evaluation, but the evidence remains limited by dataset size and scope. The current run uses 457 labeled TCGA-THCA cases and a 69-case held-out test set, so the measured differences should still be treated as preliminary.

The 69-case test set still has limited statistical power to distinguish very small AUC gains at the 95% level. With 34 positive and 35 negative test cases, the benchmark is more informative than the earlier small-cohort setup but still too small to support strong claims from changes on the order of a few thousandths of AUC.

The mapping in Eq. (11) depends on a variance estimate σ^2 from a chosen uncertainty quantification method [21], [22]. Different estimators change c even when p is fixed.

A. Knowledge Engineering Cost

The method can require a proposition inventory and a knowledge base with truth values when it is deployed with explicit symbolic rules. That requirement shifts effort from model training to knowledge engineering. The present TCGA-THCA and WiDS branches each use a deliberately small four-rule rule base, which is enough to demonstrate that the revision pathway is active but far from a comprehensive clinical ontology. The reported results should therefore not be read as evidence that these exact hand-selected rules are sufficient for clinical deployment. A fuller clinical deployment would need an independently curated rule base, broader rule coverage, pre-specified truth-value assignments, and sensitivity analysis for the initial (f_{sym}, c_{sym}) values.

This also creates an inter-rater problem. If two clinicians disagree on the appropriate truth value for a thyroid-cancer rule, the system has no automatic way to decide which prior

is correct. In practice, such disagreements would need either consensus review, sensitivity analysis over rule weights, or explicit reporting of alternative rule sets.

The present experiment also used a single operating threshold and one stratified split for each dataset. There is no external validation cohort in the current study. That is acceptable for a first methods paper, but it is a major clinical-informatics limitation: stronger claims would require testing on temporally or institutionally independent cohorts, preferably with locked preprocessing, locked rule definitions, and no retuning on the external cohort.

The current repository snapshot now includes multiple MAF files and supports the intended top-50-gene mutation panel through the downloader-integrated preprocessing path. Even so, the genomic coverage is still bounded by what has been downloaded locally, so future work should verify that the same results hold under larger cohort-level mutation exports and external validation cohorts.

VIII. Discussion and Future Work

In addition to the empirical run reported in Sec. V, the interface remains straightforward to validate more broadly in a reproducible way. The next stage of evaluation should test (i) the numerical behavior of the mappings, (ii) end-to-end predictive performance under stronger resampling and cohort variation, and (iii) clinician-facing interpretability.

A. Synthetic Sanity Checks

Synthetic settings where p and σ^2 are controlled remain useful for stress-testing the interface. For a grid such as $p \in \{0.01, 0.1, 0.5, 0.9, 0.99\}$ and $\sigma^2 \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, one can compute (f, c) using Eq. (11) and the attention reweighting using Eq. (12).

Such experiments can report tables and plots showing how c and α^{new} vary as a function of (p, σ^2) for multiple settings of γ (e.g., $\gamma \in \{0.5, 1, 2, 4\}$), and quantify how often attention mass shifts across a fixed decision threshold as σ^2 increases.

B. Dataset-Scarcity Ablation

The present study already evaluates a transformer baseline versus a transformer+NARS interface on a TCGA-THCA lymph node metastasis setting using a structured clinical benchmark. A fuller follow-up evaluation should expand this to external thyroid oncology cohorts and should report sensitivity to feature selection, target definition, and missing-data handling.

To test label-scarcity behavior more rigorously, future work should subsample the training data to multiple sizes (e.g., $n \in \{50, 100, 250, 500, 1000\}$ when feasible), repeat each setting over multiple random seeds, and report:

- **Discrimination**: AUC and accuracy.
- **Calibration**: Brier score and reliability diagrams.
- **Decision utility**: net benefit using decision-curve analysis.
- **Shift sensitivity**: an out-of-distribution split (temporal split, site split, or covariate-shifted split) to measure robustness under dataset shift.

C. Ablation of $g(f, c)$ and γ

Ablating the attention-gating function and its hyperparameters would clarify how much of the observed AUC gain comes from the specific form of the gate. A useful comparison would include:

- $g(f, c) = c^\gamma$ (this paper),
- $g(f, c) = c \cdot (1 - 2|f - 0.5|)$ (confidence with a mild "extremeness" penalty), and
- a variant where c is derived from a conjugate-Beta interpretation of (p, σ^2).

For each variant, γ should be tuned on a held-out validation set and the same discrimination, calibration, and decision-utility metrics should be reported to show how design choices in the interface affect clinician-facing reliability.

D. Error Analysis and Human-in-the-Loop Case Studies

To evaluate interpretability, future work should select 10–20 representative cases (or simulated hard cases) and present the NARS-derived explanation trace: the main propositions used, their individual (f, c) values, and how they combine into the final conclusion. This should also categorize common failure modes (variance underestimation, proposition extraction errors, and conflicting rules) and show how each failure mode manifests in the attention update.

Overall, the empirical result in this paper provides an initial proof of feasibility, while the experiments in Sec. VIII remain the next systematic validation steps needed to determine whether the observed calibration gains, threshold-behavior differences, and inference-time instrumentation benefits are robust across cohorts and clinical settings.

IX. Conclusion

This paper defined a bidirectional interface between transformers and NARS that converts neural predictive variance into NARS truth values and uses those explicit confidences to modulate transformer attention. The core contribution is an auditable, human-in-the-loop reasoning engine that mitigates the clinical black-box problem. Rather than emitting overconfident scalar risk scores on missing data, the architecture exposes what it does not confidently know and provides a direct insertion point for clinicians to inject overriding physiological rules. This effectively transforms the transformer's attention matrix from an opaque mapping into a transparent reasoning path.

Concretely, the interface (a) maps (p, σ^2) to (f, c) so that confidence decreases as epistemic variance increases, and (b) revises feature-level neural truths with explicit symbolic rules before applying a normalized attention update via a tunable confidence gain. The strongest empirical result is not a dramatic AUC jump, but the calibration-versus-discrimination profile on WiDS: the NARS-gated variant remains competitive on AUC (0.8802 versus 0.8803 for the baseline transformer) while achieving a superior expected calibration error (ECE = 0.00583 versus 0.00750 for the baseline transformer) and the best Brier score (0.05647). That is the architectural breakthrough defended in this manuscript. TCGA-THCA supports

a different claim and should be read as a multi-modal proof of concept showing that the same mathematical interface remains operational when fusing clinical variables with a genomic mutation matrix. On that 69-case held-out split, the transformer family outperformed the tested random forest, but the flat-confidence control was marginally best among transformer variants, so the result does not justify a strong superiority claim for NARS gating on its own. WiDS is therefore the primary empirical validation, while TCGA verifies modality robustness. Across both tasks, the symbolic layer also functions as an instrumentation layer: during WiDS inference, explicit ICU rules fired 13,031 times across 8,551 of 13,757 held-out stays, making the revision path physically active and human-auditable. The repository entrypoint `python main.py --run-all` executes both dataset branches sequentially and writes complete dataset-scoped metrics, traces, plots, and an aggregate run summary under `results/`. The present result should therefore be read as evidence that the symbolic uncertainty interface is operational across modalities and that its strongest empirical payoff appears in a more trustworthy style of clinical inference: calibrated probabilities, traceable rule intervention, and transparent oversight on a large high-missingness ICU benchmark.

Future work should evaluate the interface under controlled data-scarcity and dataset-shift regimes, reporting discrimination, calibration, and decision-utility metrics, and comparing against stronger uncertainty-aware baselines such as gradient-boosted trees and ensemble-based tabular predictors that do not use symbolic aggregation. Future work should also repeat the analysis with broader TCGA-THCA mutation coverage so the intended larger genomic feature panel can be assessed directly, and should quantify knowledge-base construction effort and characterize failures due to mis-specified rules, conflicting evidence, and proposition extraction errors.

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Conflicts of Interest

The author declares no conflicts of interest.

Ethics Statement

This work is methodological and did not involve new studies with human participants or animals, nor the collection of patient data; therefore ethics approval and informed consent were not required.

Data and Code Availability

The code for the implemented pipeline is available in the public repository https://github.com/ArjunCodess/NGTA. The current repository state supports both the TCGA-THCA benchmark derived from the NCI Genomic Data Commons clinical tables and MAF files, and the WiDS Datathon 2020 ICU benchmark distributed as `wids_icu.csv`. The repository entrypoint `python main.py --run-all` executes both dataset branches sequentially and writes dataset-scoped metrics, traces, plots, and an aggregate run summary; no new patient data were collected.

Author Biography

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References

[1] J. Ma *et al.*, "Improving rare disease classification using imperfect knowledge graph," *BMC Medical Genomics*, vol. 12, no. Suppl. 10, p. 195, 2019.

[2] Y. Ovadia *et al.*, "Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[3] P. Wang, "Non-axiomatic reasoning system: Exploring the essence of intelligence," Ph.D. dissertation, Indiana University, 1995.

[4] ——, *Rigid Flexibility: The Logic of Intelligence*. Springer, 2006.

[5] P. Wang and S. Awan, "Reasoning in non-axiomatic logic: A case study in medical diagnosis," in *Proceedings of the Conference on Artificial General Intelligence*, 2013, pp. 185–190.

[6] L. Rasmy *et al.*, "Med-BERT: Pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction," *npj Digital Medicine*, vol. 4, no. 1, p. 86, 2021.

[7] S. Azizi *et al.*, "Big self-supervised models advance medical image classification," in *Proceedings of the IEEE International Conference on Computer Vision*, 2021, pp. 3478–3488.

[8] F. Li *et al.*, "Fine-tuning bidirectional encoder representations from transformers (BERT)-based models on large-scale electronic health record notes: An empirical study," *JMIR Medical Informatics*, vol. 7, no. 3, p. e14830, 2019.

[9] T. Goncalves *et al.*, "A survey on attention mechanisms for medical applications: Are we moving toward better algorithms?" *IEEE Access*, vol. 10, pp. 98 975–99 003, 2022.

[10] M. Ghassemi *et al.*, "Review of challenges and opportunities for machine learning in healthcare," *BMJ*, vol. 372, p. n4, 2021.

[11] A. d'Avila Garcez *et al.*, "Neurosymbolic AI: The 3rd wave," *Artificial Intelligence*, vol. 294, p. 103504, 2020.

[12] G. Marra *et al.*, "Neural-symbolic computing: An effective methodology for principled integration of machine learning and reasoning," *Journal of Applied Logic*, vol. 6, 2021.

[13] B. P. Bhuyan *et al.*, "Neuro-symbolic artificial intelligence: A survey," *Neural Computing and Applications*, vol. 36, no. 12, pp. 6397–6423, 2024.

[14] B. C. Colelough and W. Regli, "Neuro-symbolic AI in 2024: A systematic review," arXiv:2501.05435, 2025.

[15] M. Hilario, "An overview of strategies for neurosymbolic integration," in *Connectionist-Symbolic Integration*. Routledge, 2013, pp. 15–34.

[16] U. Nawaz *et al.*, *A Review of Neuro-Symbolic AI Integrating Reasoning and Learning for Advanced Cognitive Systems*, ser. Intelligent Systems Reference Library. Springer, 2025.

[17] L. Dai *et al.*, "Automated classification of clinical diagnoses in electronic health records using transformer," *PLoS One*, vol. 20, no. 1, p. e0329963, 2025.

[18] B. Ghoshal *et al.*, "Estimating uncertainty in deep learning for reporting confidence to clinicians in medical image segmentation and diseases detection," *Computational Intelligence*, vol. 37, no. 4, pp. 1439–1460, 2021.

[19] A. Abad *et al.*, "Uncertainty quantification for machine learning in healthcare: A survey," arXiv:2505.02874, 2025.

[20] C. Blundell *et al.*, "Weight uncertainty in neural networks," in *Proceedings of the International Conference on Machine Learning*, 2015, pp. 1613–1622.

[21] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in *Proceedings of the International Conference on Machine Learning*, 2016, pp. 1050–1059.

[22] B. Lakshminarayanan *et al.*, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[23] P. Wang, "Intelligence: From definition to design," in *Proceedings of the Conference on Artificial General Intelligence*, 2022, pp. 1–12.

[24] X. Wu *et al.*, "Medical knowledge graph: Data sources, construction, reasoning, and applications," *Big Data Mining and Analytics*, vol. 6, no. 2, pp. 201–217, 2023.

[25] B. Abu-Salih *et al.*, "Healthcare knowledge graph construction: A systematic review of the state-of-the-art, open issues, and opportunities," *Journal of Big Data*, vol. 10, no. 1, p. 83, 2023.

[26] E. Rajabi and S. Kafaie, "Knowledge graphs and explainable AI in healthcare," *Information*, vol. 13, no. 10, p. 459, 2022.

[27] P. Jiang *et al.*, "Reasoning-enhanced healthcare predictions with knowledge graph community retrieval," arXiv:2410.04585, 2024.

[28] Y. Chudasama *et al.*, "Towards interpretable hybrid AI: Integrating knowledge graphs and symbolic reasoning in medicine," *IEEE Access*, vol. 13, pp. 12 345–12 360, 2025.